INCH-POUND
MS18283B(SH)
4 April 2019
SUPERSEDING
MS18283A(SH)
1 August 1968

MILITARY SPECIFICATION SHEET

RELIEF VALVE OPERATING CHARACTERISTICS VERSUS MAXIMUM OPERATING PRESSURE FOR STEAM AND GAS SERVICE

Reinstated after 4 April 2019 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The maximum operating pressure curve shall be used to identify the relationship the designer needs to consider regarding the differential between the nominal set pressure of the valve and the operating pressure of the vessel/system (see figure 1). The curve was established to distinguish the margin between operating pressure and the relief valve settings such as accumulation, blowdown (reseat), and nominal set pressure. It is used to minimize operating difficulty within the system between the control valve and the relief valve by providing an adequate differential for each steam and gas service application to stay within.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4820

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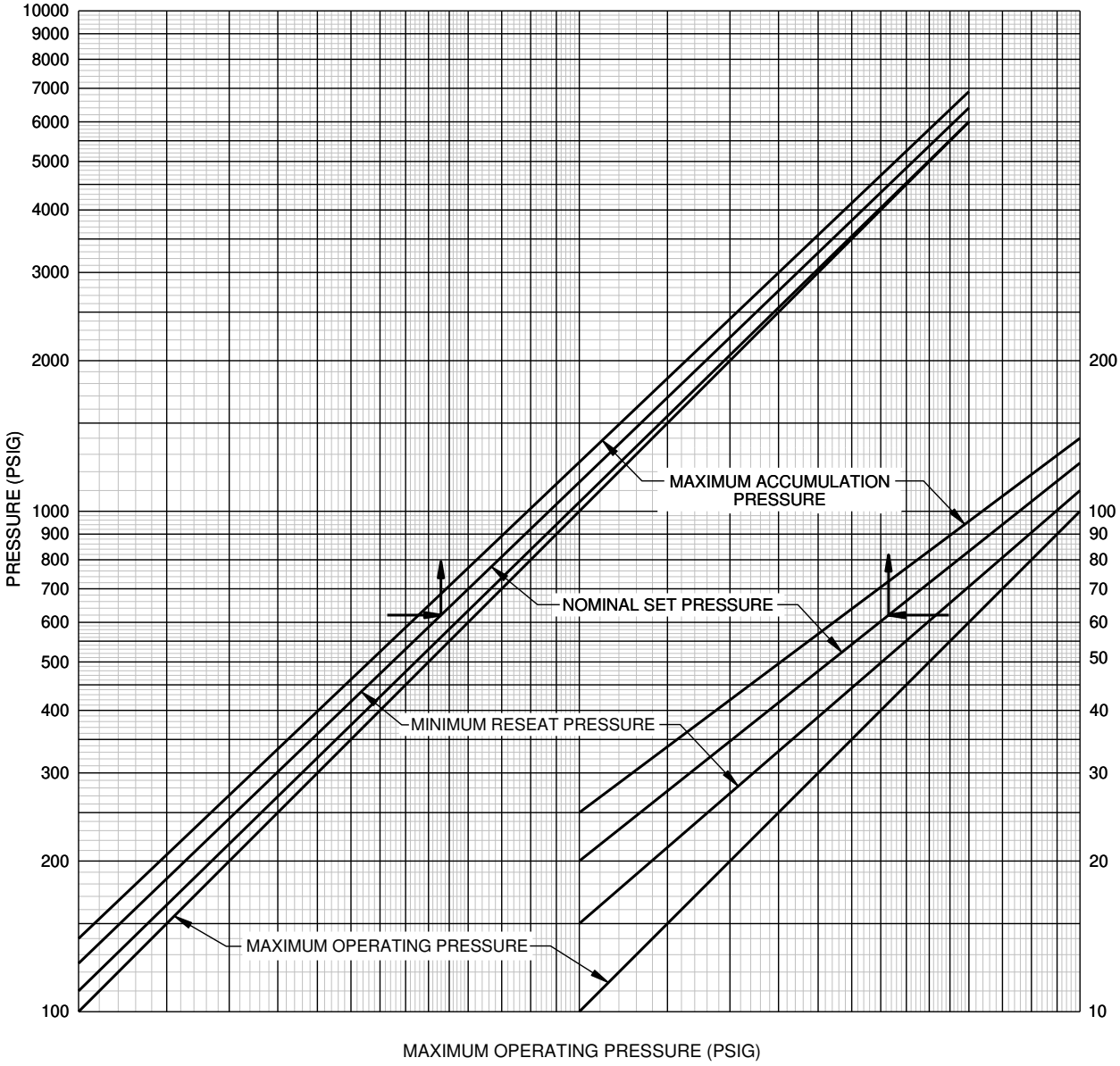


FIGURE 1. Relief valve characteristics.

NOTE: Figure 1 is used in the following way:

1. Find the desired set pressure on the left vertical ordinate (for set pressures from 100 pounds per square inch gage [PSIG] to 6500 PSIG) or on the right vertical ordinate (for set pressures from 20 PSIG to 100 PSIG). The user may extrapolate for values below 20 PSIG.

2. Find the point on the graph where the set pressure intersects the line called “NOMINAL SET PRESSURE”.

3. Moving vertically up from the point of intersection, identify the “MAXIMUM ACCUMULATION PRESSURE”.

4. Moving vertically down from the point of intersection, find the “MINIMUM RESEAT PRESSURE”.

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5. For the purposes of this specification, “Maximum Operating Pressure” is synonymous with “Maximum Allowable Operating Pressure” and “Design Pressure”, which is the pressure set limit for a system or component based on a standard wall thickness, diameter, allowable stress and safety factor. “Maximum Operating Pressure” is not synonymous with “Maximum Allowable Working Pressure”, which is case specific and based on actual system or component dimensions and may be higher than the “Maximum Operating Pressure”.

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4820-2018-028)

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